WP 05. Biomanufacturing Process Analytical Technology

RESEARCH STATUS	Proposed study. No experimental results have been claimed.
CLUSTER	Bioprocess and Life Sciences
EXCEL CROSSWALK	Excel rank 60: Bioreactor optimization
CLOSEST PROJECT	paceval_Patient_Specific_Pharmacokinetic_Rate_Kernels

Abstract

Bioprocess teams can use explicit, versioned mathematical kinetic and oxygen uptake submodels for monitoring and design-space analysis. This study defines a falsifiable and independently testable mathematical kernel for the paceval. Mathematical Engine. The workload and the evaluation criteria below are experimental plans rather than measurements.

1. Industrial scope and relationship to existing research

Distinctive contribution: Bioreactor growth and oxygen-transfer monitoring, not patient drug-rate calculations.

Workbook relationship: Excel rank 60: Bioreactor optimization. The workbook index and potential commercial benefit are subjective estimates, not measured commercial evidence.

2. Candidate mathematical kernels

Kernel 1

```
Growth: mu = mu_max*S/(Ks+S)*DO/(Ko+DO)
```

Kernel 2

```
Substrate consumption: rS = mu*X/Yxs
```

Kernel 3

```
Oxygen uptake: OUR = qO2*X
```

Kernel 4

```
Dissolved-oxygen slope: dDO_dt = kLa*(DO_star-DO)-OUR
```

Model expressions are mathematical specifications, not syntax-checked SDK strings. Implementation must declare units, parameter domains, formula IDs, regularization, model assumptions and reference tests.

3. How paceval is actually used

The host application owns parsing and dimensional validation of inputs, formula versioning, candidate generation, dataset provenance, graph-handle lifetime, independent solvers, aggregation and output. paceval owns every supported explicit scalar or composite mathematical expression and its repeated evaluation. Expression creation is performed once per stable model version. The host reuses each computation handle for all compatible input rows. Batch evaluation is used only where it preserves semantics and actually improves measured throughput. The selected paceval SDK or MCP interface must be identified in the experiment log; a local SDK call is not reported as a remote MCP benchmark.

C++ orchestrates candidate enumeration, fitted coefficients, domain-specific numerical solving, independent checking and I/O. paceval should own all supported scalar, composite cost, sensitivity and constraint formulas. Stable models use persistent computation handles and batch execution when beneficial.

4. Proposed numerical experiment and comparators

Replay 1e6 timestamped synthetic sensor snapshots, including fed-batch phase changes and controlled missing data. Add a licensed open bioprocess dataset as a second phase.

Baselines: Compare with native C++ and NumPy kinetics and host-integrated ODE predictions. Track numerical consistency, rule changeability and total monitoring latency.

Collect median, p95 and p99 latency across repeated runs; throughput, peak RAM, graph creation time, time per reused evaluation, energy where measured and complete pipeline wall time. Record hardware, compiler, precision, parallelism, batching, runtime version, warmup, data layout and input-sequence checksum. Use identical mathematical expressions, tolerances and unit handling in every baseline. Baseline alternatives should include optimized native C++ and, where domain-relevant, specialist solvers. Invalidate claims of engine-wide speedup if the measured advantage exists only in a small formula stage.

5. Independent validation and edge cases

1. Test nonnegative state and saturation limits, zero-oxygen limit and instrument drift.

2. Compare mass balance residual and oxygen balance predictions on holdout runs.

3. Use independent unit checks in g/L, h and 1/h, including conversions.

Predeclare the accepted absolute, relative and application-scaled errors. Verify each expression against an independent implementation using the same inputs and conventions. Include singular, limiting, invalid and near-boundary cases. If a formal interval claim is made, demonstrate coverage of an entire input interval using a method that actually bounds the interval; pointwise TINC checks alone are not such a proof.

6. Go or no-go assessment

A pilot is a technical go only if it passes the independently stated numerical and application acceptance tests and demonstrates either a repeatable improvement of at least 15% in full-pipeline latency or energy under matched conditions, or at least 25% reduction in measured formula-update/integration effort without a performance or quality regression. These are proposed decision thresholds, not measured outcomes. A result below the threshold is a no-go for the claimed benefit, even if the mathematical engine functions correctly.

Principal risk: Not a GMP-validated PAT system, bioreactor controller, pharmacological recommendation or a demonstration of real bioprocess data fitting.

7. Required release artifacts

Provide C++ sources, runtime version, formula manifest, dataset or synthetic generator with fixed seeds, native and high-precision references, failed-case logs, confidence intervals for timings and complete pipeline resource measurements. Never claim deployment, customer validation or certification without separate evidence.

8. Selected external references

1. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/pat-framework-innovative-pharmaceutical-development-manufacturing-and-quality-assurance

2. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q13-continuous-manufacturing-drug-substances-and-drug-products

paceval architecture: https://paceval.com/developers/

Context sources do not certify any paceval results.